A Low-Cost EEG and Head-Motion Seizure Alarm: Code-Faithful Evaluation and an Embedded Multimodal Detector

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***Abstract*—Low-cost biopotential amplifiers and microcontrollers make home seizure alarms easy to build, but their shipped detection code is rarely tested against annotated seizures. We describe an open-source wearable that combines a single-channel EEG front-end (EXG BioAmp Pill) with an MPU9250 head-motion sensor on an ESP32, and we evaluate its firmware by *code-faithful replay*: public recordings are converted to the device's signals and fed to the unmodified analyzer and a line-by-line firmware port. On CHB-MIT (23 subjects, exact FP1–FP2 montage), the shipped detectors were near chance, while a delayed-median ratio rule on the firmware's existing EEG deviation (DEV-BR) detected 52.1% of seizures at 0.97 false alarms per hour (FA/h) in held-out subjects and was unaffected by amplifier gain, clipping and sampling-rate errors. On SeizeIT2 (125 patients, 11,626 h, 883 focal-onset seizures, behind-the-ear EEG and a head-borne accelerometer), the same rule applied to EEG and to motion raised detection at the shipped firmware's alarm rate from 57.0% to 68.7% ($p<10^{-4}$). Requiring EEG *and* motion to rise together detected 76.4% (95% CI 63–88%) of focal-to-bilateral tonic–clonic seizures at 0.48 false alarms per day, versus 30.9% for EEG alone and 23.6% for motion alone at the same rate. The detectors need under 7 KB of RAM; the v5 firmware compiles for the ESP32 and reproduces the evaluation's alarms exactly. Non-convulsive focal seizures remain hard to detect, and the device has not been tested on people.**

***Index Terms*—epilepsy, seizure detection, wearable EEG, accelerometer, multimodal sensing, consumer health, ESP32, embedded signal processing**

I. Introduction

Epilepsy affects about 50 million people [1]. Unwitnessed nocturnal tonic–clonic seizures are the main risk factor for sudden unexpected death in epilepsy [2], and patient diaries miss many events [3]. Clinically endorsed wearables detect tonic–clonic seizures from movement, muscle or autonomic signals, reaching 90–96% sensitivity at 0.2–0.67 false alarms per day [4]–[6]. Focal seizures without convulsions are visible mainly in the EEG.

Open biopotential front-ends [7] and Wi-Fi microcontrollers [8] make a do-it-yourself EEG alarm cheap, but validation is weak: thresholds are tuned by hand and algorithms are re-implemented for evaluation instead of tested as shipped. We built such a device, added a head-motion sensor, and asked how well its code detects annotated seizures and what the smallest effective change is. Contributions:

1) **A built multimodal prototype**: forehead EEG and an MPU9250 inertial sensor on an ESP32, with local alarms and an offline Wi-Fi app interface (Sec. III).
2) **Code-faithful replay** of the shipped analyzer and firmware on two public datasets, with subject-level validation and empirical chance levels (Sec. IV).
3) **Embedded detectors**: a delayed-median ratio rule applied to EEG (DEV-BR) and to head acceleration (MOV-BR), in a general mode (EEG *or* motion) and a convulsive mode (EEG *and* motion), in under 7 KB of RAM and verified against the evaluation code (Sec. V).
4) **Results**: a hardware-robust EEG improvement on the exact device montage, and a co-activation alarm that reaches the false-alarm range of clinical wearables for convulsive seizures (Sec. VI).

II. Related Work

Scalp-EEG seizure detection dates to Gotman [9]; line-length is a cheap onset feature [10], and normalizing it against a running memory of past EEG changes performance [11], [12]. Baseline normalization is therefore not new; we apply it to signals an existing low-cost firmware already computes, under embedded constraints, and evaluate it against that firmware's own code. Wrist accelerometers detect generalized tonic–clonic seizures prospectively [5]. On SeizeIT2, a machine-learning fusion of behind-the-ear EEG and accelerometry detected 95.5% of tonic–clonic seizures at 1.1 false alarms per day [13]. Reduced montages retain useful but lower sensitivity [14]. Event-based scoring with tolerances is now recommended [15], [16].

III. The Monitoring System

Hardware. One differential forehead channel is amplified by the BioAmp Pill, buffered by an LM358 and digitized by the ESP32; an MPU9250 accelerometer/gyroscope mounted on the headband is read over I^2C (Fig. 1, Table I). The device runs from a USB battery bank without galvanic isolation. The

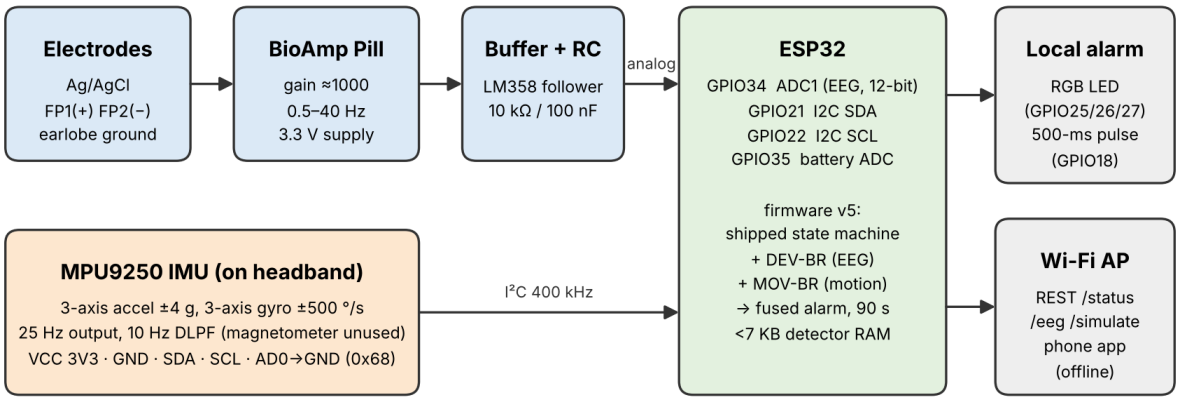


Fig. 1. Prototype (firmware v5). One forehead EEG channel (BioAmp Pill, LM358 follower, RC filter) feeds the ESP32 ADC; an MPU9250 on the headband is read over I^2C at 25 Hz. Gain and pass-band are nominal values from the documentation, not measurements.

TABLE I
IMPLEMENTATION STATUS AND SOURCE OF EACH SPECIFICATION.

Item	Specification	Status / source
EEG front-end	Ag/AgCl FP1+/FP2−; BioAmp Pill (gain ≈1000, 0.5–40 Hz, 3.3 V); LM358 follower; RC 159 Hz	built; nominal values; vendor documents 5-V use [7]
Motion sensor	MPU9250, ±4 g, ±500 °/s, 10-Hz DLPF, 25 Hz, I^2C GPIO21/22 [17]	built; not bench-characterized
ADC, sampling	12-bit, default attenuation, `delay(5)` loop	firmware v5 reports measured rate
Outputs	RGB LED, 500-ms GPIO pulse, Wi-Fi AP REST API, phone app	firmware; app not evaluated
Detectors	v4: state machine + 3-feature analyzer; v5: DEV-BR, MOV-BR, co-activation	v4 replay-evaluated; v5 compiled (792 KB flash), bit-exact offline
Bench / human data	noise, battery, timing, wear	not measured

LM358 is not rail-to-rail: at 3.3 V its output tops out near 1.8 V, about 150 mV above mid-supply [18].

Shipped detectors (v4). The firmware tracks a baseline $b[n]{=}0.99\,b[n{-}1]{+}0.01\,x[n]$ and deviation $d[n]{=}|x[n]-\lfloor b[n]\rfloor|$, with an exponentially weighted mean μ and spread σ (forgetting factor 0.995 per sample). Every 30 s it sets BROWNOUT and BLACKOUT thresholds to $\mu{+}2.5\sigma$ and $\mu{+}4\sigma$, clamped to fixed ADC ranges, and leaky counters confirm states. A Python analyzer computes line-length (LL), delta/alpha ratio and 30–45-Hz energy over 2.56 s and requires two of three features above adaptive thresholds on 7 of 8 ticks.

IV. Code-Faithful Replay

CHB-MIT [19], [20] provides the device's exact EEG montage: FP1–FP2 is reconstructed as the signed sum of the bipolar chain from FP1-F7 to FP2-F8 (160 recordings, 194 seizures, 220 h, 23 subjects; chb01 and chb21 are one subject). It has no motion data. **SeizeIT2** [21] (CC0) provides 11,626 h from 125 patients with focal epilepsy recorded in five monitoring units with a wearable: two behind-the-ear EEG channels at 256 Hz and accelerometer/gyroscope data at 25 Hz from a sensor on the back of the neck, the head-borne location closest to our headband. It contains 883 annotated seizures (55 focal-to-bilateral tonic–clonic, FBTC; 371 focal motor; 324 focal non-motor; 133 other). We use the single behind-the-ear channel of each recording (left or right side) as the device's EEG input; 386 of 2850 recordings lack head-motion data, and there motion detectors cannot fire.

Front-end model and replay. EEG passes a causal model of the nominal pass-band (0.5-Hz high-pass, 40-Hz low-pass), is resampled to 200 Hz, scaled by the nominal gain around 1.65 V and quantized to 12 bits. The analyzer is imported unmodified and driven by a simulated serial port and clock; the firmware `loop()` is ported line by line. Motion features are computed exactly as in firmware v5 (Sec. V). This is a software replay; no hardware-in-the-loop experiment was performed.

Scoring. Following SzCORE [16], alarms within 90 s merge, a seizure counts as detected if an alarm falls in [onset−30 s, offset+60 s], and other alarms are false alarms (FA). Parameters are selected by leave-one-subject-out cross-validation (LOSO): on all other subjects we pick the most sensitive setting whose pooled FA/h meets a target and apply it to the held-out subject. CIs use a 2000-resample subject bootstrap; detectors are compared per subject with the Wilcoxon signed-rank test, and seizure-level detections with an exact McNemar test. Chance levels come from 200–1000 random circular shifts of each alarm train.

V. Detectors

Why the shipped detectors fail. The firmware's spread estimate forgets within about 1 s, so 4σ excursions are routine in ordinary EEG; the analyzer's vote waits on two features that carry no seizure information (delta/alpha AUC 0.51, high-frequency energy 0.40); and fixed floors in ADC units make detection gain-dependent.

DEV-BR and MOV-BR. Let $\delta(t)$ be the per-second mean of the firmware deviation d, and $\alpha(t)$ the per-second mean

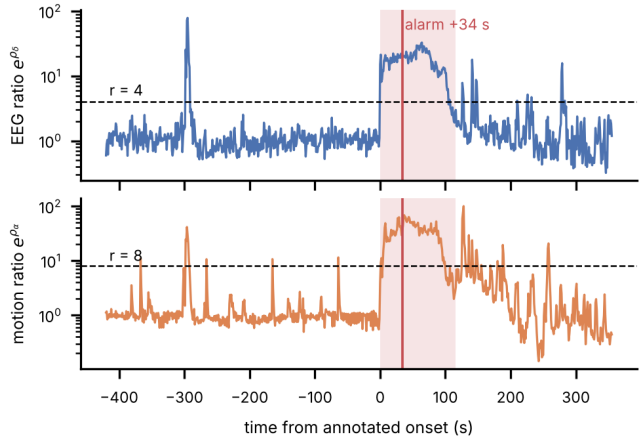


Fig. 2. Convulsive-mode alarm on an FBTC seizure (shaded) in SeizeIT2 patient 115, arising from sleep. Brief EEG artifacts and isolated movements cross one threshold but not both; the seizure raises both ratios and the alarm fires after the 30-s persistence. Shown: the detected sleep FBTC seizure with the shortest latency and no false alarm within 30 min (illustrative, not typical).

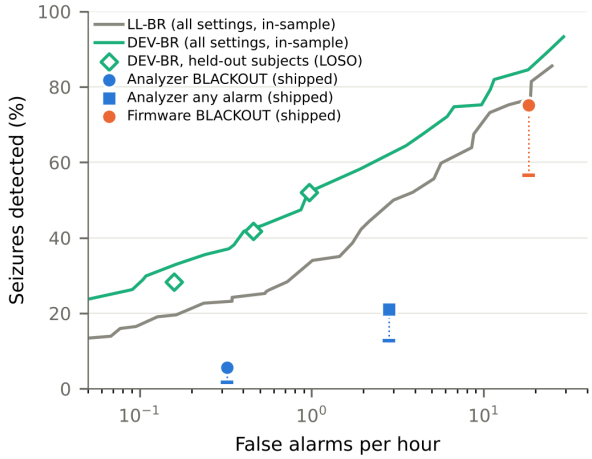


Fig. 3. CHB-MIT, FP1–FP2: sensitivity versus FA/h. Lines: best in-sample ratio trade-offs; diamonds: DEV-BR on held-out subjects; markers: shipped detectors, with dashes at their chance levels.

of $\|\mathbf{a} - \mathrm{EMA}_{1\,\mathrm{s}}(\mathbf{a})\|$ for the 25-Hz acceleration $\mathbf{a}$. For either signal s,

$$\rho_s(t) = \log \frac{s(t)}{\mathrm{median}\{s(\tau) : t{-}329 \le \tau \le t{-}30\}}, \quad (1)$$

so each wearer is compared with their own background 30–330 s earlier and the units of the sensor cancel. A condition holds when $\rho \ge \log r$ for D consecutive seconds; MOV-BR may also require that 2–7 Hz power (Hann window, last 4 s) exceed a share q, as in clonic jerking. The *general mode* raises an event when the EEG *or* the motion condition starts; the *convulsive mode* raises an alarm when $\rho_\delta \ge \log r_E$ *and* $\rho_\alpha \ge \log r_M$ hold together for D seconds. Each output fires at most once per 90 s and not before 330 s.

Firmware v5. The general event drives the existing BROWNOUT warning and the convulsive alarm drives BLACKOUT (red LED, GPIO pulse, app); the v4 serial format is unchanged. The detectors hold two 330-value histories and use `std::nth_element` once per second (5 KB), plus 1.6 KB for motion features; v5 compiles for the ESP32 at 792 KB flash (v4: 767 KB). Its parameters are the modal LOSO choices: general mode $r{=}2$, $D{=}20$ s for EEG, $r{=}24$, $q{=}0.4$, $D{=}20$ s for motion (106/125 folds); convulsive mode $r_E{=}4$, $r_M{=}8$, $D{=}30$ s (122/125 folds). On the host, the C++ code matched the Python motion features to float precision and reproduced every alarm of the evaluation (2850 of 2850 SeizeIT2 recordings identical, 11,743 alarms). It has not yet been timed on the ESP32.

VI. Results

CHB-MIT (device montage, EEG only). The analyzer's BLACKOUT alarm detected 5.7% at 0.32 FA/h; the firmware reached 75.3% at 18.29 FA/h, but randomly shifted copies of its alarms already reach 56.6% (Table II). DEV-BR detected 101/194 seizures (52.1%, CI 36–74%) at 0.97 FA/h with a median latency of 22.0 s (Fig. 3), about 7 times the analyzer at a comparable alarm rate. It was better in 21 of 23 subjects and worse in none ($p{=}6{\times}10^{-5}$), against a chance level of 6.5%. Without normalization the same procedure reached 32.5%, and logistic regression and boosted trees on 14 normalized features did not do better (best 51.0% at $\le$1 FA/h). With parameters frozen, DEV-BR stayed at 50.0%–52.1% and 0.93–0.98 FA/h across gains of 500–2000, LM358 clipping, a 3.1-V ADC range and a 6-ms loop, while the firmware moved between 45.9% and 86.1% at 7.36–24.37 FA/h (Table IV; simulation).

SeizeIT2: shipped firmware and general mode. On behind-the-ear EEG the shipped firmware detected 57.0% at 7.01 FA/h (168.2 per day), against a chance level of 29.9% (Table III). At the same alarm rate, EEG-or-motion detected 68.7% (53 patients better, 21 worse, $p{=}4{\times}10^{-5}$) and DEV-BR alone 63.2%. At $\le$1 FA/h, DEV-BR detected 33.1% (macro 47.6%; chance 4.7%), MOV-BR 28.3%, and their combination 31.1%: motion detected seizures the EEG missed but added as many false alarms, so the general mode did not beat EEG alone ($p{=}0.059$). Gyroscope magnitude was weaker than acceleration (21.0%), and a rhythm gate was selected in every MOV-BR fold. Detection depended strongly on seizure type (Fig. 4a): 96.4% of FBTC seizures, 36.7% of focal motor and 15.7% of focal non-motor seizures.

SeizeIT2: convulsive mode. Requiring EEG and head motion to rise together removed most false alarms from ordinary movement and from EEG artifacts. With parameters chosen for FBTC seizures at $\le$0.02 FA/h, co-activation detected 42/55 FBTC seizures in 37 patients (76.4%, CI 63–88%) at 0.48 false alarms per day, including 20 of 25 that arose from sleep (Fig. 2); 69 of 125 patients had no false alarm. At the same rate, EEG alone detected 30.9% and motion alone 23.6% (co-activation found 26 FBTC seizures that EEG missed and missed 1, $p{=}4{\times}10^{-7}$; versus motion, 30 and 1, $p{=}3{\times}10^{-8}$; Fig. 4b). With a 0.05-FA/h target it detected 83.6% (CI 71–94%) at 0.91 false alarms per day. The median latency was 72 s (IQR 50–97) from the annotated focal onset, which precedes

TABLE II

CHB-MIT, FP1–FP2 (23 SUBJECTS, 194 SEIZURES, 220 H; NOMINAL FRONT-END). BRACKETS: SUBJECT-BOOTSTRAP 95% CI. "MACRO": MEAN OF PER-SUBJECT SENSITIVITIES. "CHANCE": SENSITIVITY OF RANDOMLY CIRCULARLY SHIFTED ALARMS. BR ROWS: HELD-OUT SUBJECTS (LOSO).

Detector	Detected	Sensitivity [95% CI]	Macro	FA/h	Latency (s)	Chance
Analyzer, BLACKOUT (shipped)	11/194	5.7% [2–12]	10.0%	0.32	26.3	1.6%
Analyzer, any alarm (shipped)	41/194	21.1% [14–29]	25.4%	2.82	28.4	12.8%
Firmware, BLACKOUT (shipped)	146/194	75.3% [60–94]	87.6%	18.29	14.5	56.6%
LL-BR, $\leq$1 FA/h	49/194	25.3% [15–42]	40.1%	1.04	32.0	4.8%
FUSE-BR, $\leq$1 FA/h	78/194	40.2% [24–64]	56.0%	0.96	21.5	–
DEV-BR, $\leq$1 FA/h	101/194	52.1% [36–74]	68.6%	0.97	22.0	6.5%
DEV-BR, $\leq$0.5 FA/h	81/194	41.8% [26–65]	59.5%	0.46	23.0	3.9%
DEV-BR, $\leq$0.2 FA/h	55/194	28.4% [15–49]	46.1%	0.16	29.0	–

TABLE III

SEIZEIT2 (125 PATIENTS, 883 SEIZURES, 11,626 H; BEHIND-THE-EAR EEG THROUGH THE DEVICE MODEL, HEAD-BORNE ACCELEROMETER). ALL NON-FIRMWARE ROWS: HELD-OUT PATIENTS (LOSO). CLASS COLUMNS GIVE THE SHARE OF SEIZURES OF THAT CLASS DETECTED; "SLEEP": SEIZURES ANNOTATED AS ARISING FROM SLEEP.

Detector	Detected	Sensitivity [95% CI]	FA/h	FBTC	Focal motor	Focal non-motor	Sleep	Chance
Shipped firmware (BLACKOUT)	503/883	57.0% [47–65]	7.01	87.3%	55.3%	54.6%	64.0%	29.9%
Same false-alarm rate as the shipped firmware								
EEG: DEV-BR	558/883	63.2% [54–72]	5.73	94.5%	74.7%	47.8%	70.8%	–
EEG or motion	607/883	68.7% [57–79]	6.28	98.2%	80.1%	53.4%	83.1%	–
General mode, $\leq$1 FA/h target								
EEG: DEV-BR	292/883	33.1% [23–44]	0.93	92.7%	41.8%	16.0%	32.2%	4.7%
Motion: MOV-BR	250/883	28.3% [22–36]	1.00	90.9%	39.1%	10.8%	28.3%	4.3%
EEG or motion	275/883	31.1% [22–42]	0.97	96.4%	36.7%	15.7%	32.2%	4.7%
Convulsive mode, chosen for FBTC at $\leq$0.02 FA/h ($\approx$0.5/day)								
EEG only	21/883	2.4% [1–4]	0.020	30.9%	0.5%	0.0%	3.8%	0.2%
Motion only	14/883	1.6% [1–3]	0.019	23.6%	0.0%	0.3%	1.4%	0.1%
EEG AND motion	75/883	8.5% [4–15]	0.020	76.4%	5.4%	1.2%	11.2%	0.4%

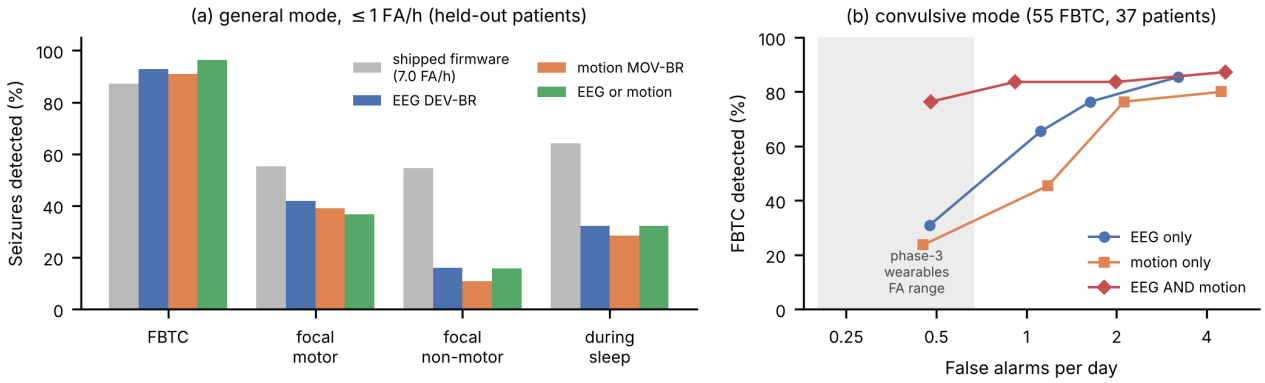


Fig. 4. SeizeIT2, held-out patients. (a) Share of seizures detected by class at $\leq$1 FA/h (the shipped firmware at its own 7.0 FA/h). (b) Convulsive mode: FBTC sensitivity versus false alarms per day for LOSO targets of 0.02–0.2 FA/h; shaded, the range reported for phase-3 wearables [4].

bilateral spread.

VII. DISCUSSION AND LIMITATIONS

Replaying the code that ships showed that the prototype's original alarms mostly fire at random, and that one rule—compare a signal with the wearer's own delayed median—fixes this for EEG and transfers to a motion sensor without knowing its units. Adding the MPU9250 changed the outcome for the seizures that matter most for safety: neither modality alone reached a usable false-alarm rate, but their co-activation detected about three in four convulsive seizures at a rate inside the range of phase-3 wearables, with under 7 KB of RAM and no machine learning. Three principles generalize: validate the shipped code, not a re-implementation; normalize against the wearer's own background so decisions do not depend on sensor units or gain; and fuse modalities with *and* when the target event moves both.

The absolute performance still falls short. Clinical wearables reach 90–96% for tonic–clonic seizures [4], and a learned multimodal model on SeizeIT2 reached 95.5% at 1.1 false alarms

TABLE IV
CHB-MIT FRONT-END ROBUSTNESS (SIMULATED). "FIXED": DEV-BR WITH r=1.25, D=20 S CHOSEN AT GAIN 1000. CLIP: LM358 1.8-V CEILING; SAT.: SATURATED SAMPLES. CELLS: SENSITIVITY / FA/H.

Condition	Sat.	DEV-BR fixed	DEV-BR LOSO	Firmware
×500	0.00%	52.1% / 0.98	47.9% / 0.94	51.5% / 7.36
×1000 (ref.)	0.00%	52.1% / 0.95	52.1% / 0.97	75.3% / 18.29
×2000	0.01%	52.1% / 0.97	50.0% / 0.96	86.1% / 24.37
×1000, LM358 clip	0.52%	51.5% / 0.97	48.5% / 0.96	73.2% / 18.01
×2000, LM358 clip	2.69%	50.0% / 0.93	50.0% / 0.93	82.0% / 23.09
×1000, 3.1-V ADC	0.00%	52.1% / 0.96	50.5% / 0.98	78.9% / 19.22
×1000, 167-Hz loop	0.00%	51.5% / 0.95	51.5% / 0.96	45.9% / 14.74

per day [13]; focal non-motor seizures were rarely detected at a usable rate. Limitations: (i) all results come from software replay; no recordings from the prototype, bench measurements of the MPU9250 or front-end, or human tests are reported. (ii) SeizeIT2 differs from the device: its EEG is behind-the-ear rather than FP1–FP2, its motion sensor sits on the neck at 25 Hz rather than on the headband, and it was recorded in hospital. (iii) The front-end model uses nominal values; non-idealities were simulated, not measured, and the BioAmp runs below its documented supply. (iv) FBTC results rest on 55 seizures from 37 patients. (v) v5 parameters are modal LOSO choices from the same data, and v5 has not been timed on the ESP32. Next steps are bench characterization, overnight recordings with the headband, and an ethics-approved study.

VIII. CONCLUSION

We built a low-cost EEG and head-motion seizure alarm and evaluated its code as shipped on two public datasets, 23 and 125 patients. The shipped detectors fail for identifiable reasons. A delayed-median ratio rule fixes the EEG detector (52.1% at 0.97 FA/h on the device montage, robust to front-end errors), and requiring EEG and motion to agree detects 76.4% of convulsive seizures at 0.48 false alarms per day, in firmware that fits in under 7 KB of RAM. Hardware, firmware, analyzer, replay benchmark and detectors are released for others to reproduce and extend.

DATA AND CODE AVAILABILITY

CHB-MIT [20] and SeizeIT2 [21] are public. Firmware (v4 and v5), analyzer, schematics, replay code and the C++ detectors are at [22].

ACKNOWLEDGMENT

The authors used an AI assistant (Claude, Anthropic) to help develop the evaluation code and firmware and to draft the manuscript; the authors reviewed all content and take full responsibility for it.

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